

Series with terms of variable sign



Outline

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Alternating series and the Leibniz test

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Introduction: series with terms of variable sign

- ▶ So far we have studied series with nonnegative terms, for which we have the comparison, limit comparison, root and ratio tests
- ▶ We now consider series whose terms may change sign: these tests do not apply directly
- ▶ We introduce **absolute convergence**, which reduces the study of $\sum a_k$ to that of the series with nonnegative terms $\sum |a_k|$
- ▶ We will prove that absolute convergence implies convergence, but the converse does not hold

Absolute convergence

Definition 1: absolutely convergent series

A series $\sum a_k$ is absolutely convergent if the series $\sum |a_k|$ converges.

Theorem 1

If a series $\sum a_k$ converges absolutely, then it converges.

Proof (1/2)

Without loss of generality, let $n_0 = 0$ and consider the following series:

$$\sum_{k=0}^{\infty} (|a_k| - a_k)$$

It is a series with nonnegative terms since, for every $k \in \mathbb{N}$, we have:

$$\begin{cases} |a_k| - a_k = -2a_k \geq 0 & \text{if } a_k < 0 \\ |a_k| - a_k = 0 & \text{if } a_k \geq 0 \end{cases}$$

Moreover, by the triangle inequality, for every $k \in \mathbb{N}$, we have:

$$\underbrace{|a_k| - a_k}_{\geq 0} = ||a_k| - a_k| \leq |a_k| + |a_k| = 2|a_k|$$

Proof (2/2)

Hence, by the comparison test for series with nonnegative terms, we have

$$\sum_{k=0}^{\infty} |a_k| \text{ convergent} \implies \sum_{k=0}^{\infty} (|a_k| - a_k) \text{ convergent}$$

Since for every $k \in \mathbb{N}$ we have $a_k = |a_k| - (|a_k| - a_k)$ and the series $\sum_{k=0}^{\infty} |a_k|$ and $\sum_{k=0}^{\infty} (|a_k| - a_k)$ converge, we have:

$$\sum_{k=0}^{\infty} a_k = \sum_{k=0}^{\infty} |a_k| - \sum_{k=0}^{\infty} (|a_k| - a_k)$$

that is, the series is the difference of two convergent series. Consequently, the series $\sum_{k=0}^{\infty} a_k$ is convergent as well. □

An alternative proof

We now see an alternative proof of the same theorem, based on splitting the partial sums into positive and negative terms.

Alternative proof (1/2)

Without loss of generality, let $n_0 = 0$ and split the sequence $\{s_n\}$ of partial sums into two sequences, the first containing only the positive terms and the second only the negative terms:

$$s_n^+ = \sum_{\substack{k \in \{0,1,\dots,n\}: \\ a_k > 0}} a_k \quad \text{and} \quad s_n^- = \sum_{\substack{k \in \{0,1,\dots,n\}: \\ a_k < 0}} -a_k$$

$$\text{hence} \quad s_n = s_n^+ - s_n^-$$

Consequently, it is enough to show that the sequences $\{s_n^+\}$ and $\{s_n^-\}$ are convergent to conclude that $\{s_n\}$ converges, and hence that the series $\sum a_k$ converges.

Alternative proof (2/2)

We observe that $\{s_n^+\}$ and $\{s_n^-\}$ are monotone non-decreasing sequences and moreover, by the triangle inequality, we have:

$$s_n^+ \leq \sum_{k=0}^n |a_k| \quad \text{and} \quad s_n^- \leq \sum_{k=0}^n |a_k|$$

On the other hand, by hypothesis the series $\sum a_k$ converges absolutely, i.e., $\sum |a_k|$ converges, and hence the quantity $\sum_{k=0}^n |a_k|$ is bounded; therefore the sequences $\{s_n^+\}$ and $\{s_n^-\}$ are bounded above and non-decreasing, and hence they converge, by the monotone sequence theorem. □

Example: absolute convergence

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k^{\alpha}} \quad \text{with } \alpha > 1$$

We have:

$$\left| \frac{(-1)^k}{k^{\alpha}} \right| = \frac{1}{k^{\alpha}}, \quad \forall k \in \mathbb{N}, k > 1$$

$$\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}} \quad \text{is convergent for } \alpha > 1$$

Therefore the series converges absolutely and hence it converges.

Absolute convergence and simple convergence

Absolute convergence implies ordinary convergence, also called simple convergence:

$$\sum |a_k| \text{ convergent} \quad \Rightarrow \quad \sum a_k \text{ convergent} \quad (1)$$

but the converse is not true (we will give a counterexample later):

$$\sum a_k \text{ convergent} \quad \nRightarrow \quad \sum |a_k| \text{ convergent} \quad (2)$$

Hence absolute convergence is a sufficient but not necessary condition for ordinary convergence.

Summary

Absolute convergence:

- ▶ $\sum a_k$ is absolutely convergent if the series with nonnegative terms $\sum |a_k|$ converges
- ▶ To study it we use the tests for series with nonnegative terms

Theorem:

- ▶ Absolute convergence $\Rightarrow$ convergence (two proofs: comparison with $\sum (|a_k| - a_k)$, or partial sums s_n^+ and s_n^-)
- ▶ The converse is false: absolute convergence is sufficient but not necessary

Alternating series and the Leibniz test

Introduction: alternating series and the Leibniz test

- ▶ Among series with terms of variable sign, a particularly simple case is given by alternating series, for which the following convergence test holds
- ▶ The **Leibniz test** allows us to establish convergence even when the series does **not** converge absolutely
- ▶ We will also see an estimate of the error made by approximating the sum with a partial sum, and several examples of application

Theorem 2: Leibniz (alternating series) test

Consider the series

$$\sum_{k=n_0}^{\infty} (-1)^k a_k \quad \text{with} \quad a_k \geq 0, \forall k$$

If the sequence $\{a_k\}$ is decreasing and $a_k \rightarrow 0$ as $k \rightarrow +\infty$, then the series is convergent. Moreover:

$$s_{2n} = \sum_{k=n_0}^{2n} (-1)^k a_k \downarrow s \quad \text{and} \quad s_{2n+1} = \sum_{k=n_0}^{2n+1} (-1)^k a_k \uparrow s$$

as $n \rightarrow +\infty$

- ▶ The partial sums with even index approximate the sum s from above and those with odd index from below.
- ▶ The Leibniz test can clearly be applied also if the terms eventually have alternating signs and the sequence $\{a_k\}$ is eventually decreasing.

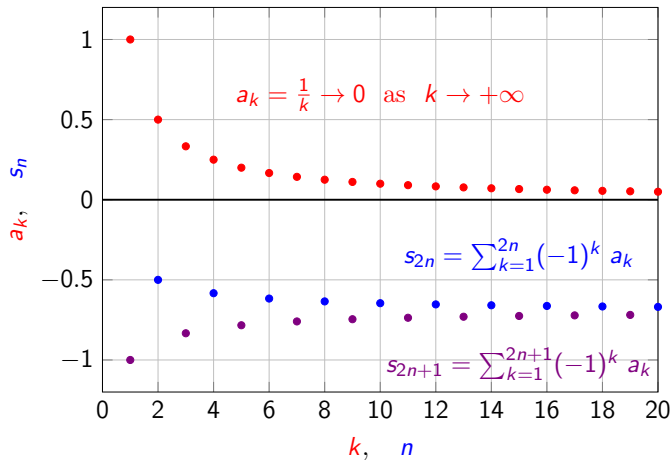
Example: Leibniz test (1/2)

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} (-1)^k \frac{1}{k}$$

The sequence $a_k = \frac{1}{k}$ is decreasing and non-negative. Moreover $\frac{1}{k} \rightarrow 0$ as $k \rightarrow +\infty$, hence it satisfies the two conditions of the Leibniz test theorem. Consequently, the series is convergent.

Example: Leibniz test (2/2)



A counterexample

The series $\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$ converges and is a counterexample to (2), that is, a series that converges but does not converge absolutely, since:

$$\sum_{k=1}^{\infty} \left| \frac{(-1)^k}{k} \right| = \sum_{k=1}^{\infty} \frac{1}{k} \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges (harmonic series)}$$

Proof of the Leibniz test (1/3)

Consider the sequence of partial sums $\{s_n\}$ with $n_0 = 0$ and the two subsequences extracted from it, $\{s_{2n}\}$ and $\{s_{2n+1}\}$.

For $\{s_{2n}\}$ we have:

$$s_0 = a_0, \quad s_2 = s_0 - a_1 + \underbrace{a_2}_{\leq a_1} \leq s_0, \quad s_4 = s_2 - a_3 + \underbrace{a_4}_{\leq a_3} \leq s_2, \quad \dots$$

hence the sequence $\{s_{2n}\}$ is monotone decreasing.

For $\{s_{2n+1}\}$ we have:

$$s_1 = a_0 - a_1, \quad s_3 = s_1 + a_2 - \underbrace{a_3}_{\leq a_2} \geq s_1, \quad s_5 = s_3 + a_4 - \underbrace{a_5}_{\leq a_4} \geq s_3, \quad \dots$$

hence the sequence $\{s_{2n+1}\}$ is monotone increasing.

Proof of the Leibniz test (2/3)

Moreover, we have:

$$s_1 \leq s_{2n+1} = s_{2n} - a_{2n+1} \leq s_{2n} \leq s_0$$

therefore $\{s_{2n+1}\}$ is bounded above and $\{s_{2n}\}$ is bounded below. The two sequences are hence convergent, by the monotone sequence theorem.

The two sequences converge to the same limit, because

$$0 \leq s_{2n} - s_{2n+1} \leq a_{2n+1} \rightarrow 0 \quad \text{as} \quad n \rightarrow +\infty$$

Proof of the Leibniz test (3/3)

Calling s this limit, since $\{s_{2n}\}$ is monotone decreasing and $\{s_{2n+1}\}$ is monotone increasing, we have:

$$s_{2n} = \sum_{k=n_0}^{2n} (-1)^k a_k \downarrow s \quad \text{and} \quad s_{2n+1} = \sum_{k=n_0}^{2n+1} (-1)^k a_k \uparrow s$$

$$\text{as } n \rightarrow +\infty$$

Hence the series is convergent since $s_{2n+1} \rightarrow s$ and $s_{2n} \rightarrow s$ as $n \rightarrow +\infty$, and consequently we have:

$$s_n \rightarrow s \quad \text{as } n \rightarrow +\infty$$



Corollary 1: of the Leibniz test theorem

Given a series satisfying the hypotheses of the Leibniz test theorem, we have:

$$\underbrace{|s - s_m|}_{=|\sum_{k=m+1}^{\infty} (-1)^k a_k|} \leq a_{m+1}, \quad \forall m \in \mathbb{N}$$

- ▶ For every m , the error made by approximating s with s_m is, in absolute value, bounded above by the value of the first omitted term. In other words, the tail of the series tends to a value less than or equal to a_{m+1} .

Proof

Since

$$s_{2n+1} \uparrow s \quad \text{and} \quad s_{2n} \downarrow s \quad \text{as } n \rightarrow +\infty$$

we therefore have, for every $n \in \mathbb{N}$:

$$s_{2n-1} \leq s \leq s_{2n} \quad \text{and} \quad s_{2n+1} \leq s \leq s_{2n}$$

from which we deduce

$$0 \leq s - s_{2n-1} \leq s_{2n} - s_{2n-1} = a_{2n}$$

$$0 \leq s_{2n} - s \leq s_{2n} - s_{2n+1} = a_{2n+1}$$

Therefore, for every m , whether even or odd, we have:

$$|s - s_m| = \left| \sum_{k=m+1}^{\infty} (-1)^k a_k \right| \leq a_{m+1}$$



Example (1/2)

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} (-1)^k \frac{k-1}{k^2+k}$$

We have:

$$a_k = \frac{k-1}{k^2+k} = \frac{k-1}{k(k+1)} \geq 0, \forall k \in \mathbb{N}, k \geq 1$$

$$\frac{k-1}{k(k+1)} \sim \frac{1}{k} \rightarrow 0 \text{ as } k \rightarrow +\infty$$

hence the series does not converge absolutely.

Example (2/2)

However, the series is decreasing for $k \geq 2$ since:

$$\underbrace{\frac{k}{(k+1)(k+2)}}_{=a_{k+1}} \leq \underbrace{\frac{k-1}{k(k+1)}}_{=a_k} \iff k^2 \leq k^2 + k - 2 \iff k \geq 2$$

hence the series converges by the Leibniz test, since it is eventually decreasing.

Example (1/2)

Let us determine the behavior of the series:

$$\sum_{k=2}^{\infty} (-1)^k \frac{\log k}{k}$$

We have:

$$a_k = \frac{\log k}{k} \geq 0, \forall k \geq 2 \quad \text{and} \quad \frac{\log k}{k} \rightarrow 0 \quad \text{as} \quad k \rightarrow +\infty$$

moreover

$$\frac{\log k}{k} > \frac{1}{k} \quad \text{for} \quad k \geq 3 \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k} \quad \text{diverges}$$

hence the series does not converge absolutely.

Example (2/2)

Proving algebraically that the sequence $\{a_k\}$ is decreasing is complicated; instead, we pass from the discrete to the continuous setting. We have, for $x \in \mathbb{R}$ and $x \geq 2$:

$$f(x) = \frac{\log x}{x} \quad \text{and} \quad f'(x) = \frac{1 - \log x}{x^2} \leq 0 \quad \text{for} \quad x \geq e$$

It follows that f is decreasing for $x \geq e$; consequently the sequence $a_k = f(k)$ is decreasing for $k \geq 3$ (the first integer $> e$).

Hence the series converges by the Leibniz test.

Sum of series

if $\sum a_k$ converges and $\sum b_k$ converges
then $\sum (a_k + b_k)$ converges

if $\sum a_k$ converges and $\sum b_k$ diverges
then $\sum (a_k + b_k)$ diverges

This can be checked by viewing the series as the limit of the sequence of partial sums and applying the theorem on the limit of a sum.

Example (1/2)

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{k + 1 + (-1)^k k^2}{k^3}$$

We split it into the sum of two series:

$$\sum_{k=1}^{\infty} \frac{k + 1 + (-1)^k k^2}{k^3} = \sum_{k=1}^{\infty} \frac{k + 1}{k^3} + \sum_{k=1}^{\infty} \frac{(-1)^k}{k}$$

Example (2/2)

For the first series we have:

$$\frac{k+1}{k^3} \geq 0, \forall k \geq 1 \quad \text{and} \quad \frac{k+1}{k^3} \rightarrow 0 \quad \text{as} \quad k \rightarrow +\infty$$

hence it converges by the limit comparison test, since:

$$\frac{k+1}{k^3} \sim \frac{1}{k^2} \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k^2} \quad \text{converges}$$

The second series converges by the Leibniz test; hence the original series converges.

Example

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} (-1)^k \left(\frac{\sqrt{k} + (-1)^k}{k} \right)$$

We split it into the sum of two series:

$$\sum_{k=1}^{\infty} (-1)^k \left(\frac{\sqrt{k} + (-1)^k}{k} \right) = \sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}} + \sum_{k=1}^{\infty} \frac{1}{k}$$

The first series converges by the Leibniz test; the second diverges (harmonic series); hence the original series diverges.

Example

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

We have:

$$\frac{(-1)^{k+1}}{k} = -\frac{(-1)^k}{k}, \quad \forall k \in \mathbb{N}, k > 1$$

moreover, the sequence $a_k = \frac{1}{k}$ is decreasing and $\frac{1}{k} \rightarrow 0$ as $k \rightarrow +\infty$, hence it satisfies the two conditions of the theorem.

Consequently, the series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$ converges.

Summary

Leibniz test:

- ▶ $\sum (-1)^k a_k$ with $a_k \geq 0$ decreasing (also eventually) and $a_k \rightarrow 0$ converges
- ▶ $s_{2n} \downarrow s$ (from above) and $s_{2n+1} \uparrow s$ (from below)
- ▶ Error estimate: $|s - s_m| \leq a_{m+1}$ (first omitted term)

Techniques in the examples:

- ▶ $\sum \frac{(-1)^k}{k}$ converges but not absolutely
- ▶ Monotonicity via algebraic inequalities or passing to the continuous setting ($f'(x) \leq 0$)
- ▶ Splitting into a sum of series: convergent + convergent, convergent + divergent

Final summary

- ▶ For series with terms of variable sign we first study **absolute convergence** with the tests for series with nonnegative terms
- ▶ Absolute convergence $\Rightarrow$ simple convergence, but not conversely (counterexample: $\sum \frac{(-1)^k}{k}$)
- ▶ For alternating series we apply the **Leibniz test**: $a_k \geq 0$, decreasing and infinitesimal
- ▶ The corollary bounds the error made by truncating the series: at most the first omitted term
- ▶ A series can be split into a sum of simpler series studied separately